



COMPASS

Clinical OUD Map & Policy Assistance Support System
· University of Tennessee · Summer 2026

150+

SOURCE DOCUMENTS

9

RESEARCH TOPICS

95

TN COUNTIES MAPPED

336

KNOX CO. DRD CASES

University of Tennessee · Knoxville
Axiom Systems Lab & College of Social Sciences
Open-access · No login required
compass.axiomssystemslab.com

THE PROBLEM

TENNESSEE'S OPIOID CRISIS

#1

Drug OD deaths nationally

3,616

TN overdose deaths in 2022

18+

% OUD patients go untreated

Critical information about OUD treatment — **clinical guidelines, state policy, local surveillance data** — is scattered across dozens of agencies, databases, and PDF reports. Clinicians, researchers, and policymakers can't easily access or synthesize it in time to act.

WHAT IS COMPASS

An AI Research Assistant for OUD

COMPASS is a **Retrieval-Augmented Generation (RAG) chatbot** that puts 150+ clinical guidelines, peer-reviewed papers, and Tennessee policy documents at your fingertips. Ask questions in plain language and get evidence-based answers with citations — in seconds.

RAG-POWERED AI

TENNESSEE-SPECIFIC

EVIDENCE-GROUNDED

FREE TO ACCESS

WHO CAN USE COMPASS

HOW IT WORKS — RAG PIPELINE

1

User Asks a Question

Type any OUD-related question in plain English — clinical, policy, or data. No special syntax needed.

E.G. "WHAT'S THE RECOMMENDED BUPRENORPHINE DOSE FOR FENTANYL-DEPENDENT PATIENTS?"

2

Vector Similarity Search

The question is encoded as a vector. FAISS searches 150+ indexed documents and retrieves the most semantically relevant passages.

FAISS · FASTEMBED · SEMANTIC SEARCH

3

Context Passed to LLM

Retrieved passages are bundled with the question and sent to the AI model — Claude (Anthropic), Ollama (local), or HuggingFace.

CLAUDE AI · OLLAMA · HUGGINGFACE

4

Grounded Answer Returned

The model generates a cited answer based only on retrieved evidence — not hallucination. Every response links to source documents you can verify.

CITATIONS INCLUDED · SOURCE-VERIFIED

RESEARCH LIBRARY — 9 TOPIC AREAS



OUD Medications & Treatment

buprenorphine · methadone · naltrexone



Clinical Practice Guidelines

ASAM · SAMHSA · CDC · WHO · VA/DoD



TN OUD Data & Surveillance

SUDORS · NSDUH · CSMD · Knox Co.



Behavioral & Counseling Treatment

CBT · motivational interviewing · CM



Neuroscience of Opioid Dependence

receptor pharmacology · neuroadaptation



Co-occurring Mental Health

dual-diagnosis · integrated treatment



Health Equity & Access Gaps in TN

racial disparities · rural barriers



Harm Reduction

naloxone · fentanyl strips · SSPs



Policy & Law

TN abatement · Purdue settlement · MOUD law

LIVE DATA & VISUALIZATIONS



Clinicians & Prescribers

Find dosing protocols, induction guides, ASAM/SAMHSA standards, and COWS scoring tools at the point of care.



Public Health Workers

Explore county-level overdose trends, treatment facility locations, and wastewater surveillance data across TN.



Researchers

Rapidly search 150+ indexed documents for evidence on MOUD efficacy, behavioral interventions, and health equity.



Policy Makers

Access TN abatement reports, federal prescribing regulations, correctional MOUD policy, and settlement data.



Social Workers

Find harm reduction strategies, co-occurring mental health resources, and housing/access gap data by county.



Students & Trainees

Learn OUD neuroscience, treatment evidence, and Tennessee policy — with sources you can verify and cite.

TECHNICAL ARCHITECTURE



React UI
Vite ·
TypeScript
Tailwind CSS



FastAPI
Python
backend
REST
endpoints



**FAISS
Index**
Vector
search
150+ docs



LLM
Claude ·
Ollama
HuggingFace

EXAMPLE QUERIES

"What are the **recommended first-line medications** for OUD in Tennessee?"

"What does *research* say about **OUD treatment during pregnancy**?"

"How do **Tennessee's buprenorphine guidelines** compare to ASAM standards?"

"Which **TN counties** have the **highest overdose death rates** in 2025?"

"What are the **CDC 2022 opioid prescribing recommendations**?"

● Treatment Facility Finder

SAMHSA-certified opioid treatment programs across all US states — filterable with contact info and certification dates.

● Knox County DRD Dashboard

336 individual medical examiner records (Jan–May 2025) with full toxicology panels and demographic breakdowns.

● TN County Overdose Map

Interactive choropleth of all 95 TN counties — overdose deaths, treatment gaps, and prescription rates by geography.

● National NSDUH Estimates

OUD prevalence, treatment access, and unmet need across all 50 states from the 2023–2024 SAMHSA small-area estimates.

● Wastewater Surveillance

TN municipal wastewater opioid biomarker data — an early warning signal for community-level use patterns.

KEY BENEFITS

- ✓ **Cited answers, not hallucination.** Every response is grounded in retrieved source documents with direct links you can verify.
- ✓ **Tennessee-first lens.** Local Knox County data, TN policy documents, and state surveillance are prioritized alongside national standards.
- ✓ **No login — fully open access.** Available to any clinician, researcher, or student at compass.axiomsystemsllab.com.
- ✓ **One tool for everything.** Chat, live data dashboards, clinical calculators (COWS, DAST), concept maps, and prescriber guides in a single interface.
- ✓ **Equity-aware.** Includes dedicated topic areas for racial/ethnic disparities, rural access barriers, and structural inequities in OUD care.

